



# Structural Organisation in Animals

In the animal kingdom you meet a large variety of organisms, both unicellular and multicellular. In **unicellular organisms**, all functions like **digestion, respiration and reproduction** are performed by a **single cell**. In the complex body of **multicellular animals** the same basic functions are carried out by **different groups of cells** in a well organised manner.

The body of a simple organism like **Hydra** is made of different types of cells, and the number of cells in each type can be in **thousands**. The **human body** is composed of **billions of cells** to perform various functions.

● **Tissue** - in multicellular animals, a **group of similar cells** along with **intercellular substances** that together perform a **specific function** (one or more functions) in the body.

You may be surprised to know that **all complex animals consist of only four basic types of tissues**. These tissues are organised in specific proportion and pattern to form an **organ** like stomach, lung, heart and kidney. When **two or more organs** perform a common function by their physical and/or chemical interaction, they together form an **organ system**, e.g., digestive system, respiratory system, etc.

**Cells, tissues, organs and organ systems** split up the work in a way that exhibits **division of labour** and contribute to the **survival of the body as a whole**.

## 7.1 Animal Tissues

The **structure of the cells vary according to their function**. Therefore, the tissues are different and are broadly classified into **four types**: (i) **Epithelial**, (ii) **Connective**, (iii) **Muscular** and (iv) **Neural**.

### 7.1.1 Epithelial Tissue

We commonly refer to an epithelial tissue as **epithelium** (pl.: epithelia). This tissue has a **free surface**, which faces either a body fluid or the outside environment and thus provides a **covering or a lining** for some part of the body. Epithelia are **sheet-like tissues** lining the body's surface and its **cavities, ducts and tubes**.

The cells are **compactly packed with little intercellular matrix**. There are **two types** of epithelial tissues namely **simple epithelium** and **compound epithelium**. **Simple epithelium** is composed of a **single layer of cells** and functions as a lining for body cavities, ducts and tubes. The **compound epithelium** consists of **two or more cell layers** and has a **protective function** as it does in our skin.

On the basis of structural modification of the cells, **simple epithelium** is further divided into **three types** (Figure 7.1): (i) **Squamous**, (ii) **Cuboidal**, (iii) **Columnar**.

| Simple epithelium | Structure                                                                                    | Location                                                 | Main function              |
|-------------------|----------------------------------------------------------------------------------------------|----------------------------------------------------------|----------------------------|
| Squamous          | Single thin layer of flattened cells with irregular boundaries                               | Walls of blood vessels and air sacs of lungs             | Forms a diffusion boundary |
| Cuboidal          | Single layer of cube-like cells                                                              | Ducts of glands and tubular parts of nephrons in kidneys | Secretion and absorption   |
| Columnar          | Single layer of tall and slender cells; nuclei at the base; free surface may have microvilli | Lining of stomach and intestine                          | Secretion and absorption   |

The epithelium of the **proximal convoluted tubule (PCT)** of the nephron in the kidney has **microvilli**.

- If the columnar or cuboidal cells bear **cilia** on their free surface they are called **ciliated epithelium** (Figure 7.1d). Their function is to **move particles or mucus in a specific direction** over the epithelium. They are mainly present in the inner surface of **hollow organs like bronchioles and fallopian tubes**.
- Some of the columnar or cuboidal cells get specialised for secretion and are called **glandular epithelium** (Figure 7.2). They are mainly of two types: **unicellular**, consisting of isolated glandular cells (**goblet cells** of the alimentary canal), and **multicellular**, consisting of a cluster of cells (**salivary gland**).

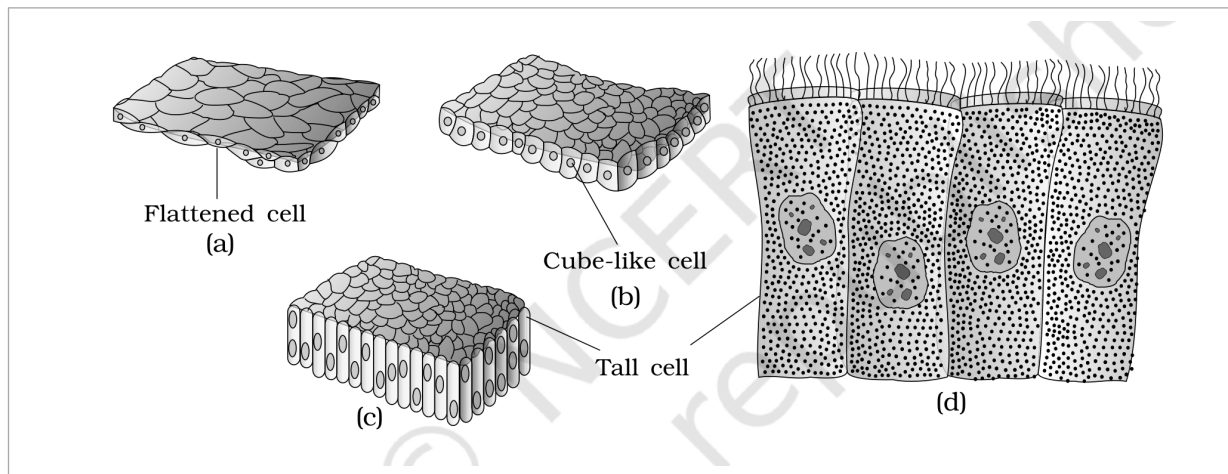


Figure 7.1 Simple epithelium: (a) Squamous (b) Cuboidal (c) Columnar (d) Columnar cells bearing cilia

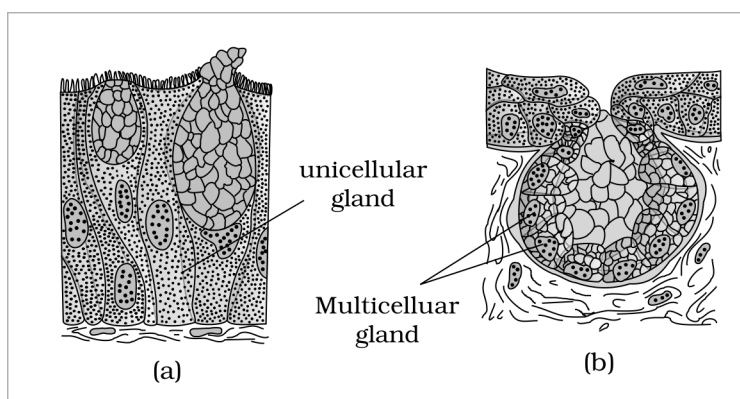


Figure 7.2 Glandular epithelium: (a) Unicellular (b) Multicellular

On the basis of the **mode of pouring of their secretions**, glands are divided into two categories namely **exocrine** and **endocrine** glands.

| Gland type | Duct                                     | Secretions                                                                  |
|------------|------------------------------------------|-----------------------------------------------------------------------------|
| Exocrine   | Products released through ducts or tubes | Mucus, saliva, earwax, oil, milk, digestive enzymes and other cell products |
| Endocrine  | Do not have ducts                        | Hormones, secreted directly into the fluid bathing the gland                |

**Compound epithelium** is made of **more than one layer** (multi-layered) of cells and thus has a **limited role in secretion and absorption** (Figure 7.3). Their main function is to **provide protection against chemical and mechanical stresses**. They cover the **dry surface of the skin**, the moist surface of **buccal cavity, pharynx**, inner lining of ducts of **salivary glands** and of **pancreatic ducts**.

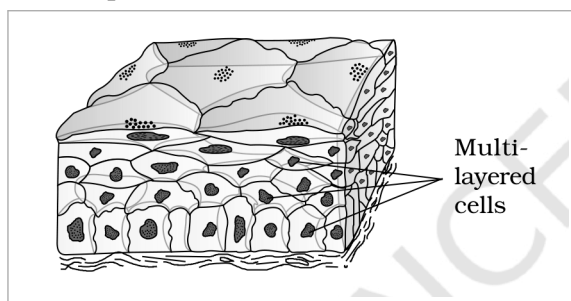


Figure 7.3 Compound epithelium

All cells in epithelium are held together with **little intercellular material**. In nearly all animal tissues, **specialised junctions** provide both **structural and functional links** between its individual cells. **Three types of cell junctions** are found in the epithelium and other tissues. These are called **tight, adhering and gap junctions**.

| Cell junction   | Function                                             |
|-----------------|------------------------------------------------------|
| Tight junctions | Help to stop substances from leaking across a tissue |

| Cell junction      | Function                                                                                                                                                                    |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Adhering junctions | Perform cementing to keep neighbouring cells together                                                                                                                       |
| Gap junctions      | Facilitate the cells to communicate with each other by connecting the cytoplasm of adjoining cells, for rapid transfer of ions, small molecules and sometimes big molecules |

7.1.2

### Connective Tissue

**Connective tissues** are **most abundant and widely distributed** in the body of complex animals. They are named connective tissues because of their special function of **linking and supporting** other tissues/organs of the body. Diverse types of connective tissues **bind together, support, strengthen, protect, and insulate** other tissue in the body. **Connective tissue covers all three** of the other tissue types.

They range from **soft connective tissues** to **specialised types**, which include **cartilage, bone, adipose, and blood**. In all connective tissues **except blood**, the cells secrete **fibres of structural proteins called collagen or elastin**. The fibres provide **strength, elasticity and flexibility** to the tissue. These cells also secrete **modified polysaccharides**, which accumulate between cells and fibres and act as **matrix (ground substance)**. In short, **soft connective tissues** consist of **protein fibres** as well as a variety of **cells arranged in a ground substance**.

Connective tissues are classified into **three types**: (i) **Loose connective tissue**, (ii) **Dense connective tissue** and (iii) **Specialised connective tissue**.

A

### Loose Connective Tissue

- **Loose connective tissue** has cells and fibres **loosely arranged in a semi-fluid ground substance**, for example, **areolar tissue** present beneath the skin (Figure 7.4). Often it serves as a **support framework for epithelium**. It contains **fibroblasts** (cells that produce and secrete fibres), **macrophages** and **mast cells**.
- **Adipose tissue** is another type of **loose connective tissue** located mainly beneath the skin. The cells of this tissue are **specialised to store fats**. The excess of nutrients which are not used immediately are **converted into fats** and are stored in this tissue. Thus adipose tissue is a **reservoir of stored energy**.

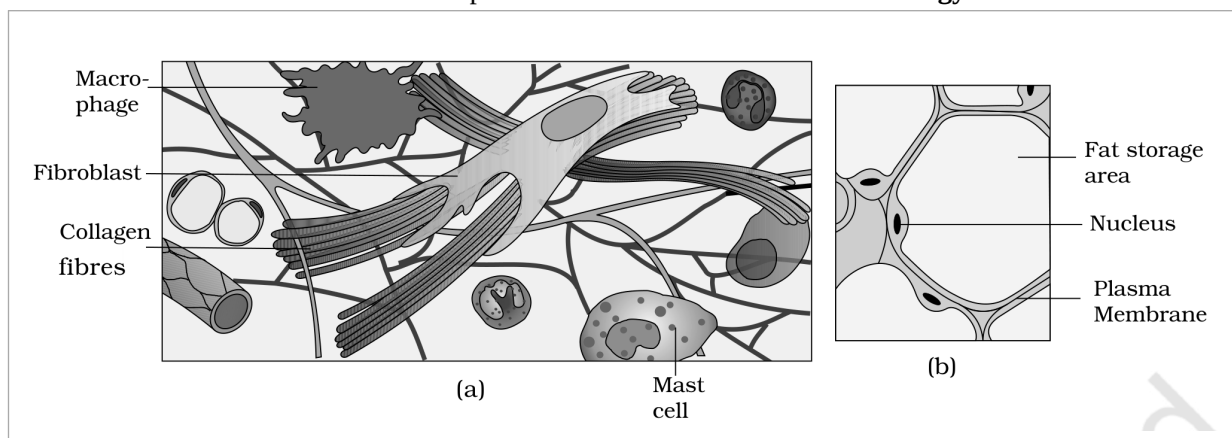


Figure 7.4 Loose connective tissue: (a) Areolar tissue (b) Adipose tissue

B

### Dense Connective Tissue

**Fibres and fibroblasts** are **compactly packed** in the dense connective tissues. Orientation of fibres shows a **regular or irregular pattern** and they are called **dense regular** and **dense irregular** tissues.

- In the **dense regular** connective tissues, the **collagen fibres** are present in rows between many parallel bundles of fibres. **Tendons**, which attach skeletal muscles to bones, and **ligaments**, which attach one bone to another, are examples of this tissue.
- **Dense irregular** connective tissue has fibroblasts and many fibres (mostly collagen) that are **oriented differently** (Figure 7.5). This tissue is present in the **skin**.

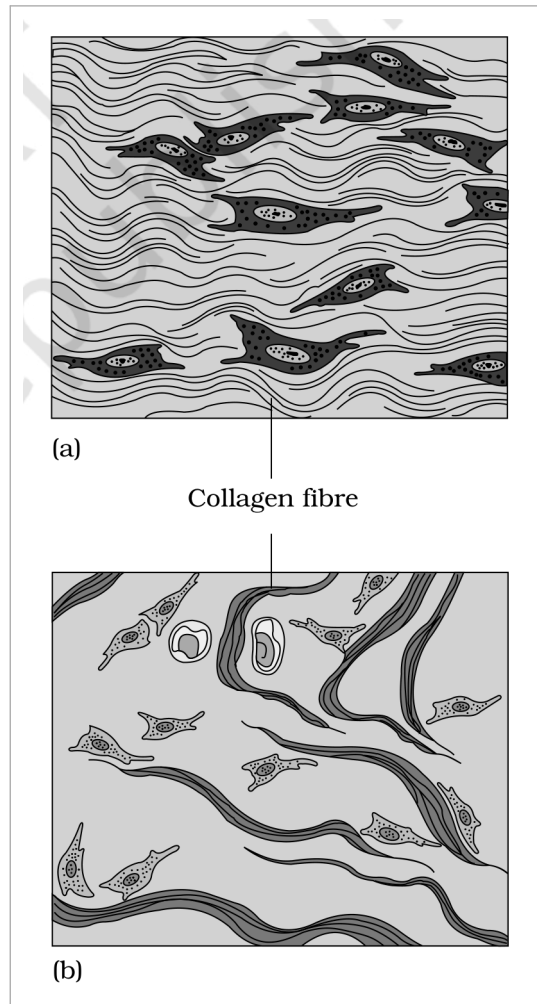


Figure 7.5 Dense connective tissue: (a) Dense regular (b) Dense irregular

## c

### Specialised Connective Tissue

**Cartilage, bones and blood** are various types of **specialised connective tissues**. (The chapter **summary** classifies **cartilage, bone, blood, and adipose tissue** together as specialised connective tissues; section 7.1.2 above first classifies adipose as a loose connective tissue. Both source statements are preserved as written.) **Cartilage and bone are both structural materials.**

- **Cartilage** - its intercellular material is **solid and pliable** and **resists compression**. Cells of this tissue (**chondrocytes**) are enclosed in **small cavities within the matrix** secreted by them (Figure 7.6a).
- Most of the cartilages in **vertebrate embryos** are **replaced by bones in adults**. In adults, cartilage is present in the **tip of nose, outer ear joints, between adjacent bones of the vertebral column, limbs and hands**.
- **Bone** - has a **hard and non-pliable ground substance rich in calcium salts and collagen fibres** which give bone its strength (Figure 7.6b). It is the **main tissue that provides structural frame to the body**.
- Bones **support and protect softer tissues and organs**. The bone cells (**osteocytes**) are present in the spaces called **lacunae**.
- **Limb bones**, such as the long bones of the legs, serve **weight-bearing functions**. They also interact with **skeletal muscles** attached to them to bring about movements. The **bone marrow** in some bones is the **site of production of blood cells**.
- **Blood** - a **fluid connective tissue** containing **plasma, red blood cells (RBC), white blood cells (WBC) and platelets** (Figure 7.6c). It is the main **circulating fluid** that helps in the **transport of various substances**.

You will learn more about blood in **Chapters 17 and 18**.

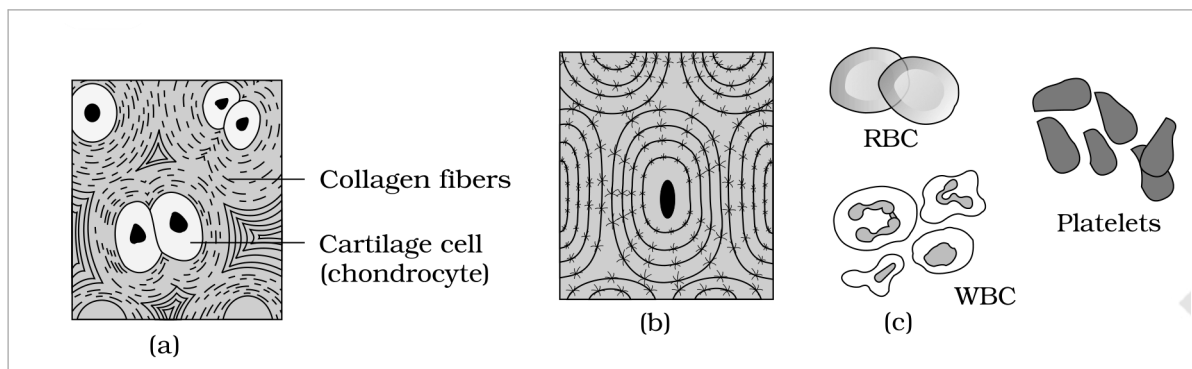


Figure 7.6 Specialised connective tissues: (a) Cartilage (b) Bone (c) Blood

7.1.3

### Muscle Tissue

Each muscle is made of many **long, cylindrical fibres** arranged in **parallel arrays**. These fibres are composed of numerous fine fibrils, called **myofibrils**. Muscle fibres **contract (shorten)** in response to stimulation, then **relax (lengthen)** and return to their uncontracted state in a coordinated fashion. Their action moves the body to **adjust to changes in the environment** and to **maintain the positions of the various parts of the body**. In general, muscles play an **active role in all the movements of the body**.

Muscles are of **three types: skeletal, smooth, and cardiac** (Figure 7.7).

| Muscle   | Striations                                          | Location                                                                                                                                                 | Control / junctions                                                                                                                 |
|----------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Skeletal | Striated (striped)                                  | Closely attached to skeletal bones; e.g., biceps - fibres bundled in a parallel fashion, several bundles enclosed by a sheath of tough connective tissue | Voluntary (learn more in Chapter 20)                                                                                                |
| Smooth   | No striations; fibres taper at both ends (fusiform) | Wall of internal organs such as blood vessels, stomach and intestine                                                                                     | Involuntary - functioning cannot be directly controlled; cell junctions hold fibres together, bundled in a connective tissue sheath |
| Cardiac  | Contractile tissue                                  | Present only in the heart                                                                                                                                | Cell junctions fuse the plasma membranes; communication junctions (intercalated discs) let the cells contract as a unit             |

**Cardiac muscle** makes up the **contractile walls of the heart**: when one cell receives a signal to contract, its neighbours are also stimulated to contract.

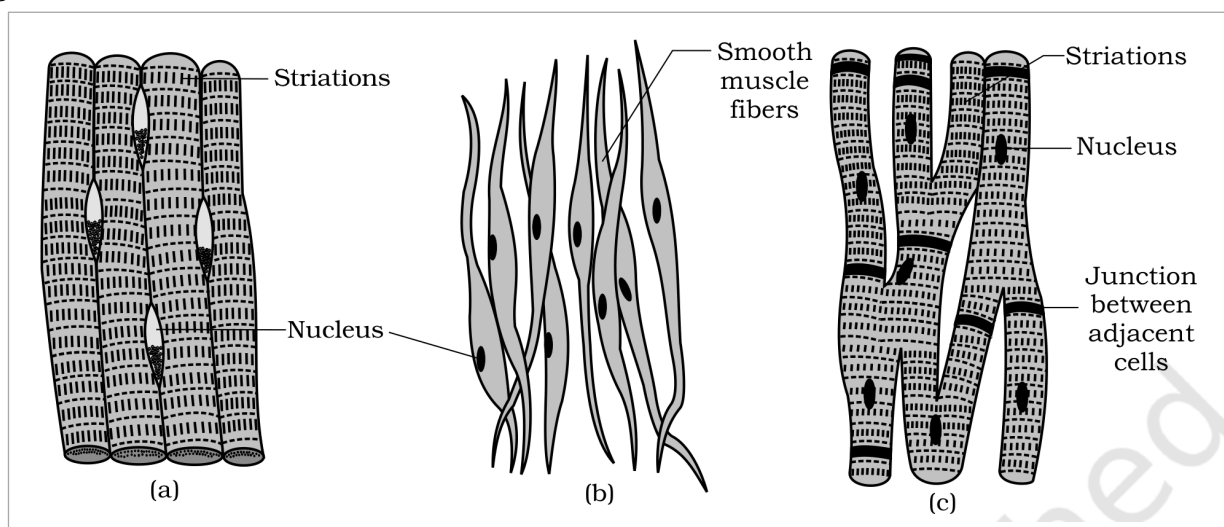


Figure 7.7 Muscle tissue: (a) Skeletal (striated) muscle tissue (b) Smooth muscle tissue (c) Cardiac muscle tissue

**Neural tissue** exerts the **greatest control** over the body's responsiveness to changing conditions. **Neurons**, the unit of the neural system, are **excitable cells** (Figure 7.8). The **neuroglial cells**, which constitute the rest of the neural system, **protect and support neurons**. Neuroglia make up **more than one-half the volume** of neural tissue in our body.

When a neuron is suitably stimulated, an **electrical disturbance** is generated which swiftly travels along its **plasma membrane**. Arrival of the disturbance at the neuron's endings, or **output zone**, triggers events that may cause **stimulation or inhibition** of adjacent neurons and other cells (details in **Chapter 21**).

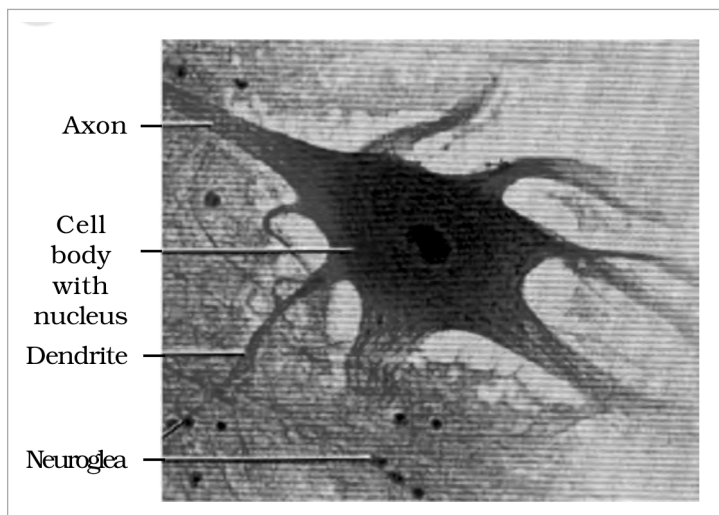


Figure 7.8 Neural tissue (Neuron with neuroglia)

The basic tissues mentioned above **organise to form organs** which in turn associate to form **organ systems** in the multicellular organisms. Such an organisation is essential for **more efficient and better coordinated activities** of millions of cells constituting an organism. **Each organ** in our body is made of **one or more type of tissues**. For example, our **heart** consists of **all the four types of tissues** - epithelial, connective, muscular and neural.

We also notice, after some careful study, that the complexity in organ and organ systems displays a certain discernable trend. This is called the **evolutionary trend** (details in class XII). You are being introduced to the **morphology and anatomy of three organisms at different evolutionary levels** to show their organisation and functioning.

- **Morphology** - the study of **form or externally visible features**. In plants or microbes, the term morphology precisely means only this; in animals it refers to the **external appearance** of the organs or parts of the body.
- **Anatomy** - conventionally used for the study of the **morphology of internal organs** in animals.

① **[NOTE]** In the requested NEET syllabus, the **Earthworm** (section 7.3) has been removed, so the chapter continues from the general organ/organ-system idea directly to the **Cockroach** and the **Frog** as the worked examples of animal organisation.

**Cockroaches** are **brown or black bodied** animals that are included in class **Insecta** of Phylum **Arthropoda**. Bright **yellow, red and green** coloured cockroaches have also been reported in tropical regions.

- Their **size** ranges from **1/4 inches to 3 inches (0.6-7.6 cm)** and they have **long antennae, legs** and a **flat extension of the upper body wall that conceals the head**.
- They are **nocturnal omnivores** that live in **damp places** throughout the world. They have become residents of human homes and thus are **serious pests and vectors of several diseases**.

The adults of the common species of cockroach, *Periplaneta americana*, are about **34-53 mm long** with **wings that extend beyond the tip of the abdomen in males**. The body of the cockroach is **segmented** and divisible into **three distinct regions - head, thorax and abdomen** (Figure 7.14).

The entire body is covered by a **hard chitinous exoskeleton** (brown in colour). In each segment, the exoskeleton has hardened plates called **sclerites** (**tergites** dorsally and **sternites** ventrally) that are joined to each other by a **thin and flexible articular membrane (arthrodial membrane)**.

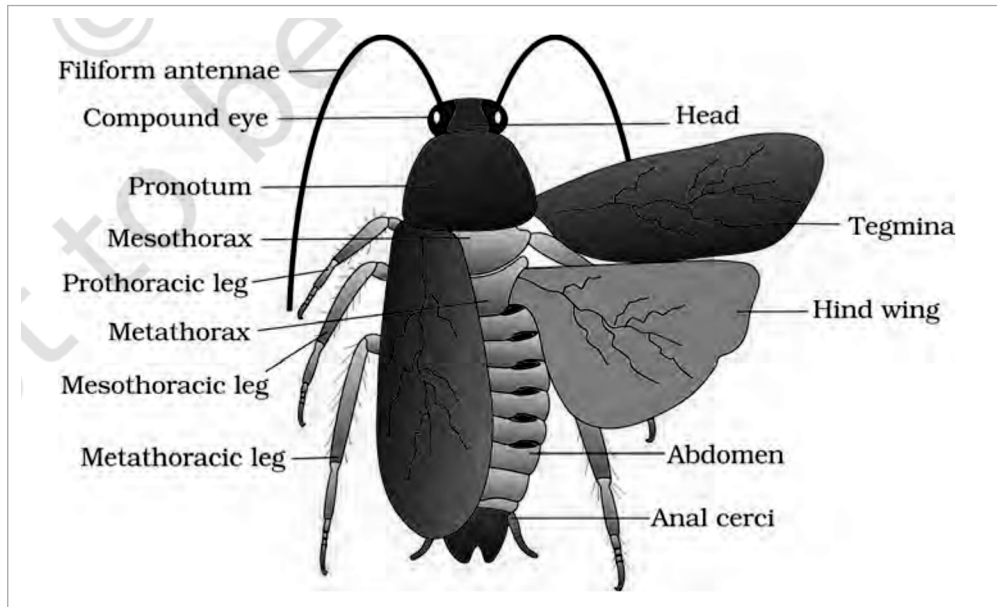


Figure 7.14 External features of cockroach

## Head

## Head

The **head** is **triangular in shape** and lies anteriorly at **right angles to the longitudinal body axis**. It is formed by the **fusion of six segments** and shows **great mobility in all directions** due to a flexible neck (Figure 7.15).

- The head capsule bears a **pair of compound eyes**. A pair of **thread-like antennae** arise from **membranous sockets** lying in front of the eyes; antennae have **sensory receptors** that help in **monitoring the environment**.
- The anterior end of the head bears appendages forming **biting and chewing type of mouthparts**, consisting of a **labrum (upper lip)**, a pair of **mandibles**, a pair of **maxillae** and a **labium (lower lip)**. A median flexible lobe, acting as a tongue (**hypopharynx**), lies within the cavity enclosed by the mouthparts (Figure 7.15b).

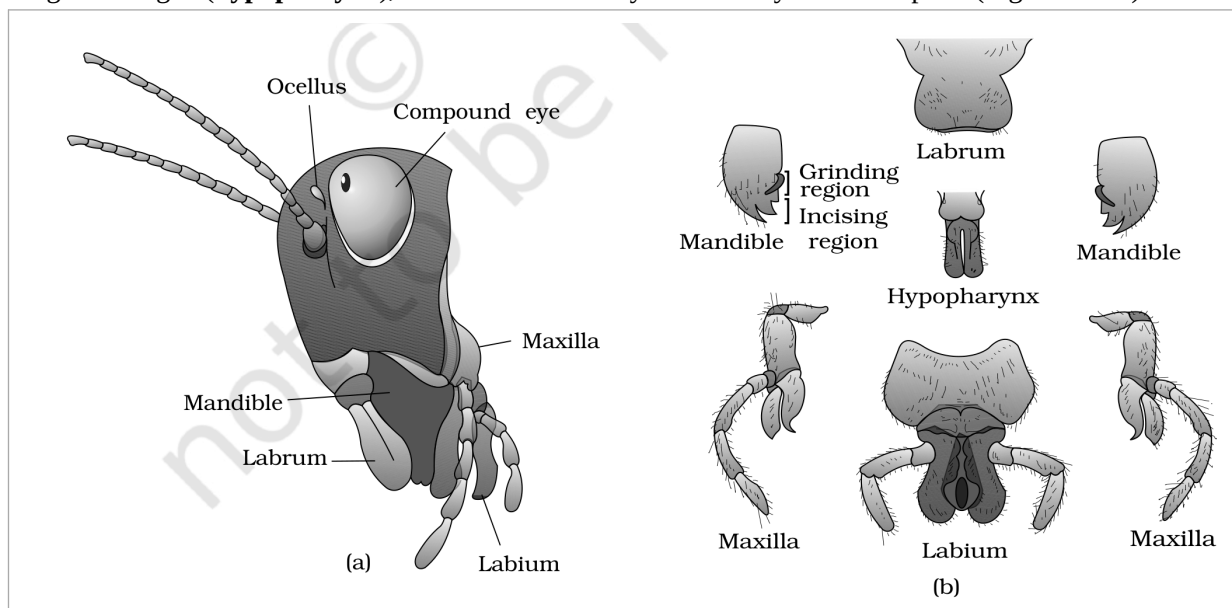


Figure 7.15 Head region of cockroach: (a) parts of head region (b) mouth parts

The **thorax** consists of **three parts - prothorax, mesothorax and metathorax**. The head is connected with the thorax by a short extension of the prothorax known as the **neck**. Each **thoracic segment bears a pair of walking legs**.

- The **first pair of wings** arises from the **mesothorax** and the **second pair** from the **metathorax**. (Thus **two pairs of wings** are present, **one pair each on the 2nd and 3rd thoracic segment**.)
- **Forewings (mesothoracic)** called **tegmina** are **opaque, dark and leathery** and cover the hind wings when at rest. The **hind wings** are **transparent, membranous** and are used in **flight**.

The **abdomen** in both males and females consists of **10 segments**.

- In **females**, the **7th sternum** is boat-shaped and together with the **8th and 9th sterna** forms a **brood or genital pouch** whose anterior part contains the **female gonopore, spermathecal pores and collateral glands**.
- In **males**, the **genital pouch or chamber** lies at the hind end of the abdomen, bounded dorsally by the **9th and 10th terga** and ventrally by the **9th sternum**. It contains the **dorsal anus, ventral male genital pore and gonapophysis**.
- Males bear a pair of short, thread-like **anal styles** which are **absent in females**. In both sexes, the **10th segment** bears a pair of jointed filamentous structures called **anal cerci**.

The **alimentary canal** present in the body cavity is divided into **three regions: foregut, midgut and hindgut** (Figure 7.16). A **pair of salivary glands** is present near the crop.

- 1 The **mouth** opens into a short tubular **pharynx**, leading to a narrow tubular passage called the **oesophagus**.
- 2 The oesophagus opens into a sac-like structure called the **crop**, used for **storing food**.
- 3 The crop is followed by the **gizzard or proventriculus**: it has an outer layer of thick circular muscles and a thick inner cuticle forming **six highly chitinous plates called teeth**, and it **helps in grinding the food particles**. The entire foregut is lined by cuticle.
- 4 A ring of **6-8 blind tubules called hepatic or gastric caeca** is present at the **junction of foregut and midgut**, which **secrete digestive juice**.
- 5 At the **junction of midgut and hindgut** is present another ring of **100-150 yellow coloured thin filamentous Malpighian tubules**, which **help in removal of excretory products from haemolymph**.
- 6 The **hindgut** is broader than the midgut and is differentiated into **ileum, colon and rectum**; the **rectum opens out through the anus**.



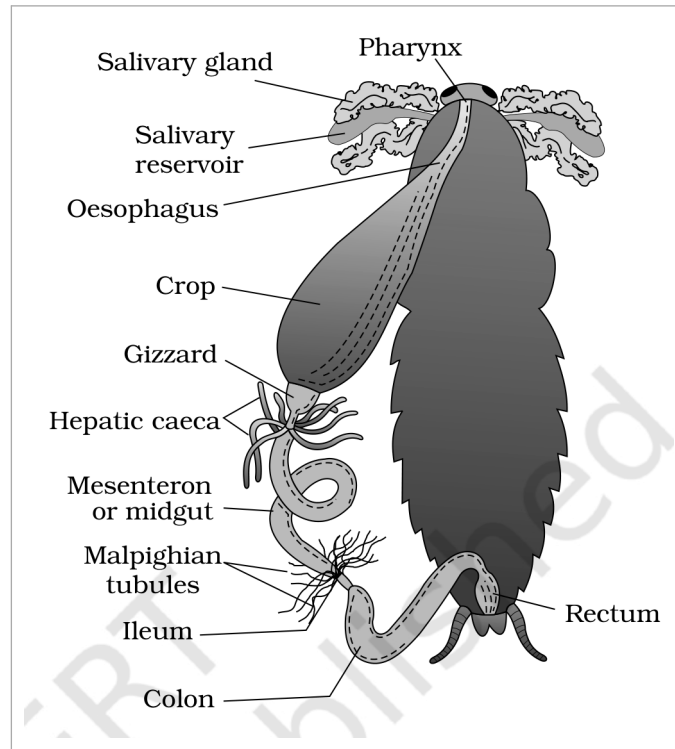


Figure 7.16 Alimentary canal of cockroach

Circ

## Circulatory System

The **blood vascular system** of cockroach is an **open type** (Figure 7.17). Blood vessels are **poorly developed** and open into a space (**haemocoel**). Visceral organs located in the haemocoel are **bathed in blood (haemolymph)**. The haemolymph is composed of **colourless plasma and haemocytes**.

The **heart** of cockroach consists of an **elongated muscular tube** lying along the **mid-dorsal line** of thorax and abdomen. It is differentiated into **funnel-shaped chambers with ostia** on either side. **Blood from sinuses enters the heart through ostia** and is **pumped anteriorly to the sinuses again**.

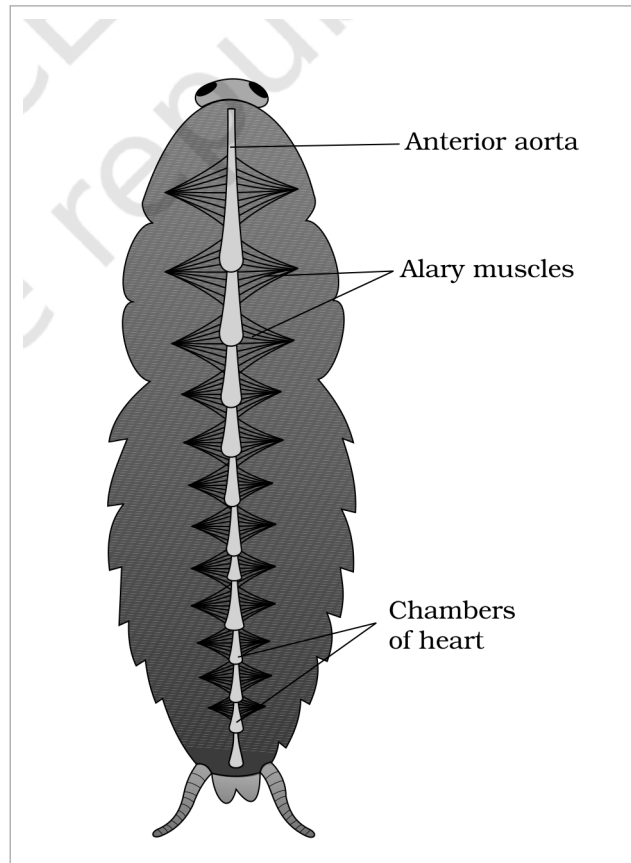


Figure 7.17 Open circulatory system of cockroach

Resp

### Respiratory System

The **respiratory system** consists of a **network of trachea**, that open through **10 pairs of small holes called spiracles** present on the lateral side of the body. **Thin branching tubes** (tracheal tubes subdivided into **tracheoles**) carry **oxygen from the air to all the parts**. The opening of the spiracles is regulated by the **sphincters**. **Exchange of gases** takes place at the tracheoles by **diffusion**.

Excr

### Excretory System

**Excretion** is performed by **Malpighian tubules**. Each tubule is lined by **glandular and ciliated cells**. They **absorb nitrogenous waste products** and **convert them into uric acid** which is excreted out through the hindgut. Therefore, this insect is called **uricotelic**. In addition, the **fat body, nephrocytes and urecose glands** also help in excretion.

Nerv

### Nervous System

The **nervous system** of cockroach consists of a series of **fused, segmentally arranged ganglia** joined by **paired longitudinal connectives** on the ventral side. **Three ganglia lie in the thorax, and six in the abdomen**. The nervous system is **spread throughout the body**: the head holds a bit of it while the rest is situated along the **ventral (belly-side) part** of the body.

① **[NOTE]** Because the nervous system is spread out, if the head of a cockroach is cut off it will still **live for as long as one week**.

In the head region, the **brain** is represented by the **supra-oesophageal ganglion** which supplies nerves to the **antennae and compound eyes**. In cockroach, the **sense organs** are **antennae, eyes, maxillary palps, labial palps, anal cerci**, etc.

The **compound eyes** are situated at the dorsal surface of the head. Each eye consists of about **2000 hexagonal ommatidia** (sing.: ommatidium). With the help of several ommatidia, a cockroach can receive **several images of an object**. This kind of vision is known as **mosaic vision**, with **more sensitivity but less resolution**, being common during night (hence called **nocturnal vision**).

Cockroaches are **dioecious** and both sexes have **well developed reproductive organs** (Figure 7.18).

- The **male reproductive system** consists of a **pair of testes**, one lying on each lateral side in the **4th-6th abdominal segments**. From each testis arises a thin **vas deferens**, which opens into the **ejaculatory duct** through the **seminal vesicle**. The ejaculatory duct opens into the **male gonopore** situated ventral to the anus.
- A characteristic **mushroom-shaped gland** is present in the **6th-7th abdominal segments** which functions as an **accessory reproductive gland**. The external genitalia are represented by **male gonapophysis or phallomere** (chitinous asymmetrical structures surrounding the male gonopore).
- The **sperms** are stored in the seminal vesicles and are glued together in the form of bundles called **spermatophores** which are discharged during copulation.
- The **female reproductive system** consists of **two large ovaries**, lying laterally in the **2nd-6th abdominal segments**. Each ovary is formed of a group of **eight ovarian tubules or ovarioles**, containing a chain of developing ova. **Oviducts** of each ovary unite into a single **median oviduct (also called vagina)** which opens into the genital chamber.
- A pair of **spermatheca** is present in the **6th segment** which opens into the genital chamber. **Sperms are transferred through spermatophores**.
- Their **fertilised eggs** are encased in capsules called **oothecae**. An ootheca is a **dark reddish to blackish brown capsule, about 3/8 inch (8 mm) long**. They are dropped or glued to a suitable surface, usually in a crack or crevice of **high relative humidity near a food source**. On an average, females produce **9-10 oothecae, each containing 14-16 eggs**.
- The development of **P. americana** is **paurometabolous**, meaning there is development through a **nymphal stage**. The **nymphs look very much like adults**. The nymph grows by **moulting about 13 times** to reach the adult form. The **next to last nymphal stage has wing pads** but only adult cockroaches have wings. (After the rupturing of a single ootheca, **sixteen young ones, called nymphs**, come out.)

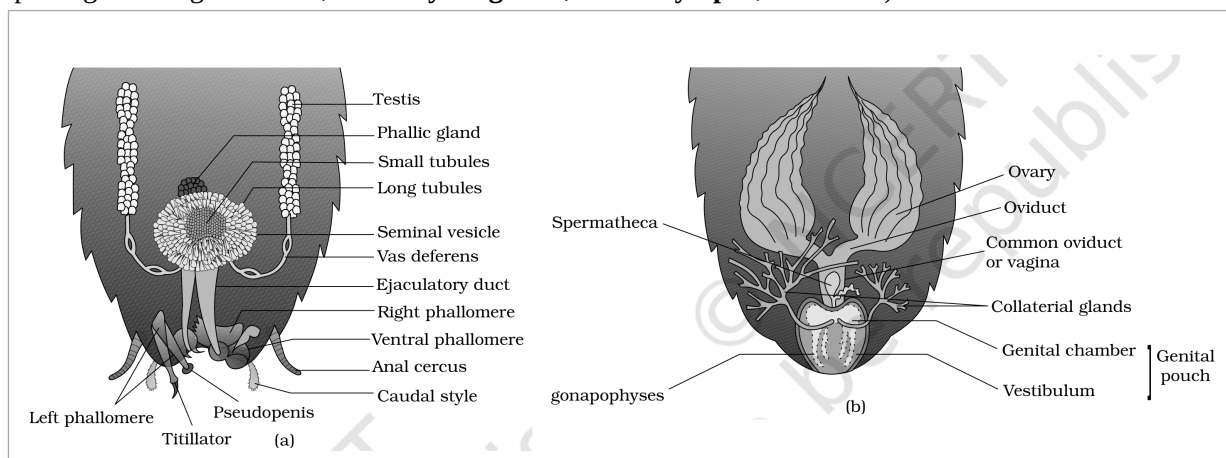


Figure 7.18 Reproductive system of cockroach: (a) male (b) female

Many species of cockroaches are **wild** and are of **no known economic importance** yet. A few species **thrive in and around human habitat**. They are **pests** because they **spoil food and contaminate it with their smelly excreta**, and they can **transmit a variety of bacterial diseases** by contaminating food material.

**Frogs** can live both on **land and in freshwater** and belong to class **Amphibia** of phylum **Chordata**. The most common species of frog found in India is **Rana tigrina**. They **do not have a constant body temperature** - their body temperature varies with the temperature of the environment; such animals are called **cold blooded or poikilotherms**.

- You might have noticed **changes in the colour** of the frogs while they are in grasses and on dry land. They have the ability to **change the colour to hide them from their enemies (camouflage)**. This **protective coloration is called mimicry**.

- Frogs are **not seen during peak summer and winter**. During this period they take shelter in **deep burrows** to protect them from extreme heat and cold. This is known as **summer sleep (aestivation)** and **winter sleep (hibernation)** respectively.

7.5.1

### Morphology

The **skin** of a frog is **smooth and slippery** due to the presence of **mucus** and is **always maintained in a moist condition**. **Mucous glands** are present in the skin, which is **highly vascularised and helps in respiration** both in water and on land. The colour of the **dorsal side** of the body is generally **olive green with dark irregular spots**, while on the **ventral side** the skin is **uniformly pale yellow**. The frog **never drinks water but absorbs it through the skin**.

The body of a frog is divisible into **head and trunk** (Figure 7.19); a **neck and tail are absent**.

- Above the mouth, a **pair of nostrils** is present. **Eyes** are bulged and covered by a **nictitating membrane** that protects them while in water. On either side of the eyes a **membranous tympanum (ear)** receives sound signals.
- The **forelimbs and hind limbs** help in **swimming, walking, leaping and burrowing**. The **hind limbs end in five digits** and are **larger and muscular** than the **fore limbs that end in four digits**. **Feet have webbed digits** that help in swimming.
- Frogs exhibit **sexual dimorphism**. **Male frogs** can be distinguished by the presence of **sound producing vocal sacs** and also a **copulatory pad on the first digit of the fore limbs**, which are **absent in female frogs**.

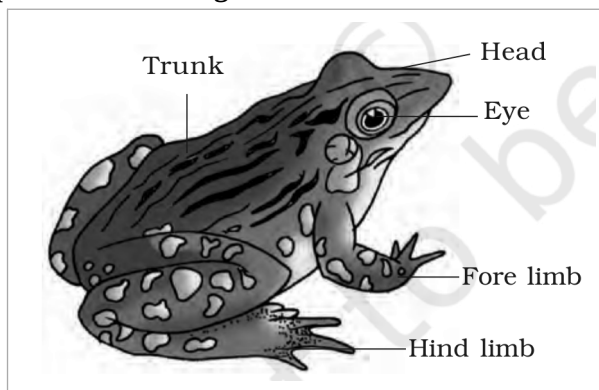


Figure 7.19 External features of frog

7.5.2

### Anatomy

The **body cavity** of frogs accommodates different organ systems - **digestive, circulatory, respiratory, nervous, excretory and reproductive** - with well developed structures and functions (Figure 7.20).

Dig

### Digestive System

The **digestive system** consists of the **alimentary canal** and **digestive glands**. The alimentary canal is **short** because frogs are **carnivores** and hence the length of the intestine is reduced. **Food is captured by the bilobed tongue**.

- 1 The **mouth** opens into the **buccal cavity** that leads to the **oesophagus** through the **pharynx**.
- 2 The oesophagus is a short tube that opens into the **stomach**. Digestion of food takes place by the action of **HCl and gastric juices** secreted from the walls of the stomach.
- 3 Partially digested food called **chyme** passes from the stomach to the first part of the small intestine, the **duodenum**, which receives **bile from the gall bladder** and **pancreatic juices from the pancreas** through a **common bile duct**.
- 4 **Bile emulsifies fat** and **pancreatic juices digest carbohydrates and proteins**; **final digestion** takes place in the intestine.
- 5 **Digested food is absorbed** by the numerous finger-like folds in the inner wall of the intestine called **villi and microvilli**.

The **undigested solid waste** moves into the **rectum** and passes out through the **cloaca**.

The main **digestive glands** are the **liver**, which secretes **bile** that is stored in the **gall bladder**, and the **pancreas**, which produces **pancreatic juice** containing digestive enzymes.

## Resp

## Respiratory System

Frogs **respire on land and in the water by two different methods**. In water, the **skin acts as the aquatic respiratory organ (cutaneous respiration)** - **dissolved oxygen** in the water is exchanged through the skin by **diffusion**. On land, the **buccal cavity, skin and lungs** act as the respiratory organs; respiration by lungs is called **pulmonary respiration**.

- The **lungs** are a pair of **elongated, pink coloured sac-like structures** present in the upper part of the trunk region (thorax). **Air enters through the nostrils** into the buccal cavity and then to the lungs. During **aestivation and hibernation**, gaseous exchange takes place through the skin.

## Circ

## Circulatory System

The **vascular system** of frog is a **well-developed closed type**. Frogs also have a **lymphatic system**. The **blood vascular system** involves **heart, blood vessels and blood**; the **lymphatic system** consists of **lymph, lymph channels and lymph nodes**. The circulatory system is **closed with single circulation**.

- The **heart** is a muscular structure situated in the upper part of the body cavity. It has **three chambers - two atria and one ventricle** - and is covered by a membrane called **pericardium**. A triangular structure called the **sinus venosus** joins the **right atrium** and receives blood through the major veins called the **vena cava**. The **ventricle** opens into a sac-like **conus arteriosus** on the ventral side of the heart.
- The blood from the heart is carried to all parts of the body by the **arteries (arterial system)**. The **veins** collect blood from different parts of the body to the heart and form the **venous system**. Special venous connections are present: between **liver and intestine** (the **hepatic portal system**) and between the **kidney and lower parts of the body** (the **renal portal system**).
- The **blood** is composed of **plasma and cells**. The blood cells are **RBC (red blood cells) or erythrocytes, WBC (white blood cells) or leucocytes and platelets**. **RBCs are nucleated** and contain the red coloured pigment **haemoglobin**.
- The **lymph** is different from blood - it **lacks few proteins and RBCs**. The blood **carries nutrients, gases and water** to the respective sites during circulation, which is achieved by the **pumping action of the muscular heart**.

## Excr

## Excretory System

The **elimination of nitrogenous wastes** is carried out by a **well developed excretory system** consisting of a pair of **kidneys, ureters, cloaca and urinary bladder**. The kidneys are **compact, dark red and bean-like structures** situated a little posteriorly in the body cavity on both sides of the vertebral column. Each kidney is composed of several structural and functional units called **uriniferous tubules or nephrons**.

- **Two ureters** emerge from the kidneys in **male frogs**; the ureters act as a **urinogenital duct** which opens into the cloaca. In **females**, the **ureters and oviduct open separately** in the cloaca. The **thin-walled urinary bladder** is present ventral to the rectum and also opens into the cloaca.
- The frog **excretes urea** and thus is a **ureotelic animal**. Excretory wastes are **carried by blood into the kidney** where they are **separated and excreted**.

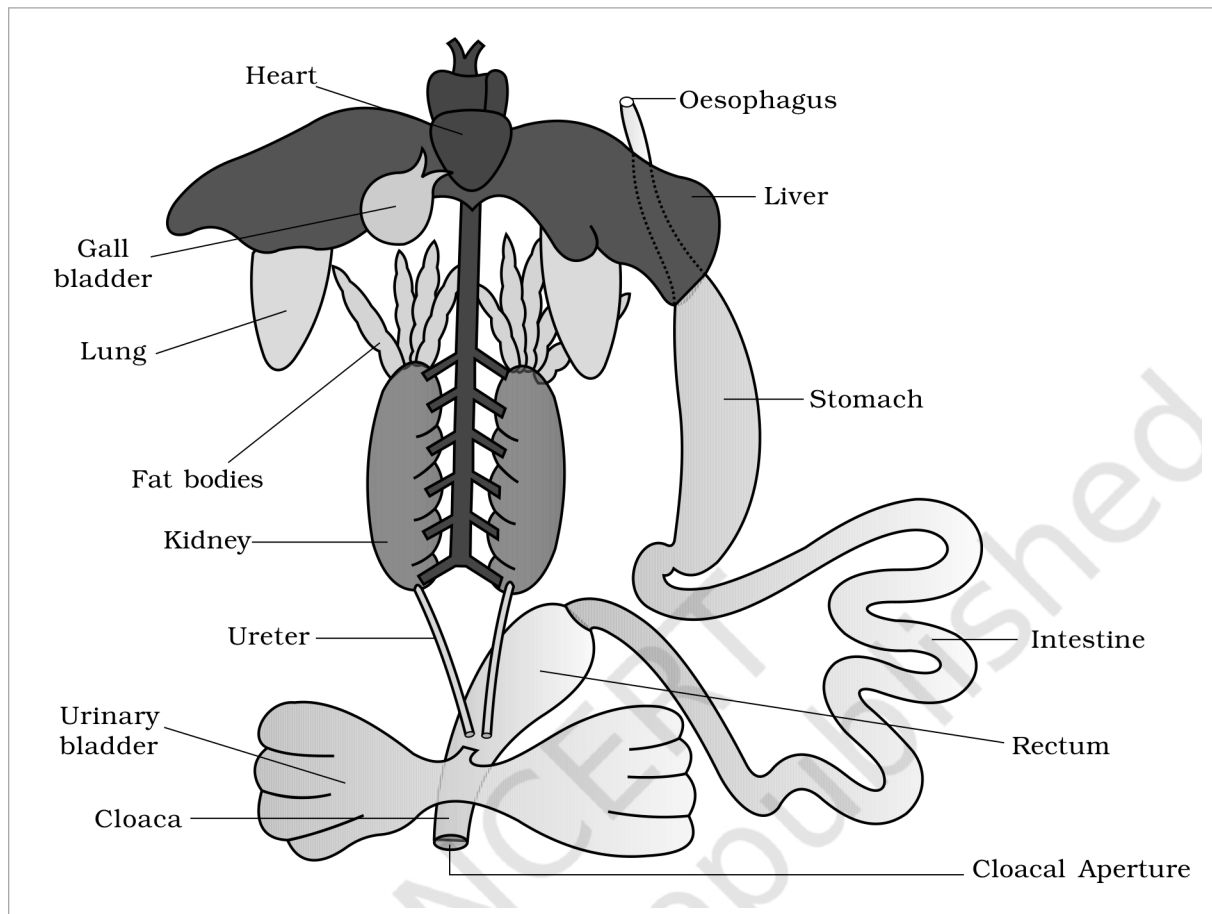


Figure 7.20 Diagrammatic representation of internal organs of frog showing complete digestive system

Ctrl

### Control and Coordination

The system for **control and coordination** is **highly evolved** in the frog and includes both the **neural system** and **endocrine glands**. The **chemical coordination** of various organs is achieved by **hormones** secreted by the **endocrine glands**. The prominent endocrine glands found in the frog are the **pituitary, thyroid, parathyroid, thymus, pineal body, pancreatic islets, adrenals and gonads**.

- The **nervous system** is organised into a **central nervous system (brain and spinal cord)**, a **peripheral nervous system (cranial and spinal nerves)** and an **autonomic nervous system (sympathetic and parasympathetic)**. There are **ten pairs of cranial nerves** arising from the brain.
- The **brain** is enclosed in a bony structure called the **brain box (cranium)** and is divided into **fore-brain, mid-brain and hind-brain**. The **forebrain** includes **olfactory lobes, paired cerebral hemispheres and unpaired diencephalon**; the **midbrain** is characterised by a **pair of optic lobes**; the **hind-brain** consists of **cerebellum and medulla oblongata**. The medulla oblongata passes out through the **foramen magnum** and continues into the **spinal cord**, which is enclosed in the vertebral column.
- The frog has different types of **sense organs**: organs of **touch (sensory papillae)**, **taste (taste buds)**, **smell (nasal epithelium)**, **vision (eyes)** and **hearing (tympanum with internal ears)**. Of these, the **eyes and internal ears** are well-organised structures and the rest are **cellular aggregations around nerve endings**.
- **Eyes** in a frog are a pair of **spherical structures** situated in the orbit in the skull; these are **simple eyes (possessing only one unit)**. **External ear is absent** in frogs and only the **tympanum** can be seen externally. The **ear** is an organ of **hearing as well as balancing (equilibrium)**.

Repr

### Reproductive System

Frogs have **well organised male and female reproductive systems**.

- **Male reproductive organs** consist of a pair of **yellowish ovoid testes** (Figure 7.21), found adhered to the upper part of the kidneys by a double fold of peritoneum called the **mesorchium**. **Vasa efferentia** are **10-12 in number** and arise from the testes; they enter the kidneys on their side and open into **Bidder's canal**. Finally it communicates with the **urinogenital duct** that comes out of the kidneys and opens into the **cloaca**.

- The **cloaca** is a small, median chamber used to pass **faecal matter, urine and sperms** to the exterior.
- The **female reproductive organs** include a pair of **ovaries** (Figure 7.22), situated near the kidneys with **no functional connection** with them. A pair of **oviducts** arising from the ovaries opens into the **cloaca separately**. A mature female can lay **2500 to 3000 ova at a time**.
- **Fertilisation is external** and takes place in **water**. Development involves a larval stage called the **tadpole**, which undergoes **metamorphosis** to form the adult.

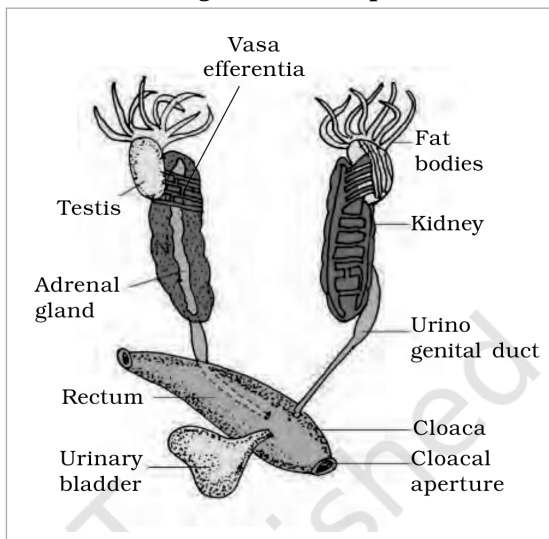


Figure 7.21 Male reproductive system

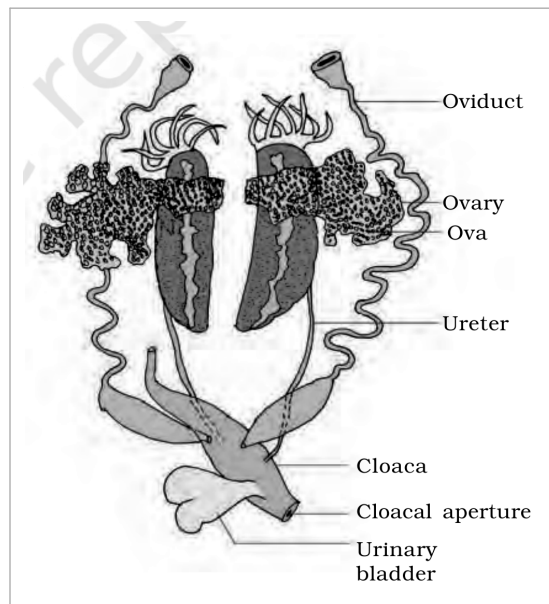


Figure 7.22 Female reproductive system

**Frogs are beneficial for mankind** because they **eat insects and protect the crop**. They **maintain ecological balance** as an important link of the **food chain and food web** in the ecosystem. In some countries the **muscular legs of frog** are used as food by man.

#### Recap

#### Quick Recap

- A **tissue** is a group of cells along with intercellular substances performing **one or more functions**. Cells, tissues, organs and organ systems exhibit **division of labour** for the survival of the whole body. Complex animals have **four tissue types**: epithelial, connective, muscular and neural.
- **Epithelia** are sheet-like tissues lining the body's surface and its **cavities, ducts and tubes**, with one **free surface**; their cells are connected at **tight, adhering and gap junctions**. Simple epithelium is squamous, cuboidal or columnar; compound epithelium is protective.
- **Connective tissues** bind together, support, strengthen, protect and insulate other tissues, and cover all three other tissue types. **Soft connective tissues** have protein fibres and varied cells in a ground substance. **Cartilage, bone, blood and adipose tissue** are specialised connective tissues; cartilage and bone are structural materials, **blood is a fluid transport tissue**, and **adipose tissue is a reservoir of stored energy**.
- **Muscle tissue** contracts in response to stimulation and helps movement. **Skeletal muscle** is attached to bones, **smooth muscle** is a component of internal organs, and **cardiac muscle** makes up the contractile walls of the heart. **Nervous tissue** exerts the greatest control over the body's responses; **neurons** are its basic units.
- The body of **Cockroach (Periplaneta americana)** is covered by a chitinous exoskeleton and divided into **head, thorax and abdomen**, whose segments bear jointed appendages. The **three thoracic segments** each bear a pair of walking legs, **two pairs of wings** are present (one pair each on the 2nd and 3rd segment), and the **abdomen has 10 segments**.
- The cockroach **alimentary canal** has a mouth with mouthparts, pharynx, oesophagus, crop, gizzard, midgut, hindgut and anus, with a **pair of salivary glands near the crop**. **Hepatic caecae** lie at the foregut-midgut junction and **Malpighian tubules** at the midgut-hindgut junction (excretion). The blood vascular system is **open**; respiration is by a network of **tracheae** opening through **spiracles**; the nervous system is **segmentally arranged ganglia with a ventral nerve cord**.
- In the cockroach, a **pair of testes** is present in the 4th-6th segments and **ovaries** in the 2nd-6th segments; **fertilisation is internal**. A female produces **9-10 oothecae** bearing developing embryos, and after rupturing of a

single ootheca **sixteen nymphs** come out.

- The **Indian bullfrog, Rana tigrina**, is covered by **skin** with mucous glands (highly vascularised, aids respiration in water and on land), and the body is divisible into **head and trunk**. A muscular **bilobed tongue** captures prey. Its alimentary canal (oesophagus, stomach, intestine, rectum) opens into the **cloaca**; the main **digestive glands are liver and pancreas**.
- The frog can respire through **skin in water and lungs on land**. Its circulatory system is **closed with single circulation** and **RBCs are nucleated**. The nervous system is organised into **central, peripheral and autonomic** parts. The urinogenital system has **kidneys and urinogenital ducts** opening into the cloaca.
- The male reproductive organ is a **pair of testes** and the female a **pair of ovaries**. A female lays **2500-3000 ova** at a time; **fertilisation and development are external**, and the eggs hatch into **tadpoles** which metamorphose into frogs.

#### Appendix

#### Terms Used in the Exercises

A few exercise questions use terms that the supplied chapter text does not itself define. Those **gap** terms are answered here so the exercises can be attempted from these notes alone; every other exercise is answerable from the sections and figures above.

- **Axon** [Q7(b)] - the **single long process of a neuron** that conducts the nerve impulse **away from the cell body**, towards the neuron's endings (output zone). It is labelled in Figure 7.8 but not defined in the running text.
- **Simple gland vs compound gland** [Q9(e)] - a **simple gland** has an **unbranched duct**, whereas a **compound gland** has a **branched duct system** draining many secretory units. (This is distinct from the unicellular/multicellular classification given in 7.1.1.)
- **Coxa** [Q10(e)] - the **basal segment of an insect leg**, which attaches the leg to the thoracic segment. (The cockroach's thoracic segments - mesothorax and metathorax - are described in 7.4.1.)
- **Protonema** [Q10(e)] - the **green, branched, filamentous first stage** of the gametophyte in the life cycle of a **moss**. It is a plant structure and does not belong to the animals studied in this chapter.
- **Typhlosole** [Q11(e)] - the **internal median fold of the dorsal wall of the intestine of the earthworm**, which **increases the absorptive surface area**. (Earthworm is outside the NEET syllabus; the term is listed only because an exercise assumes it.)